

The Minimal Polynomial, Cyclic Subspaces, and Jordan Normal Form

Math 5250 — Week 4

Motivation

If $f(x) = P_A(x)$ factors over an algebraically closed field as

$$f(x) = \prod_i (x - \lambda_i)^{n_i}$$

and all $n_i = 1$, then A is diagonalizable.

If some $n_i > 1$, the answer is open: sometimes A is diagonalizable, sometimes not.

To understand the general case, we need some theory of polynomials.

Polynomials over $F[x]$

$F[x]$: polynomials with coefficients in a field F (not assumed algebraically closed).

$$f(x) = a_0 + a_1x + \cdots + a_nx^n$$

has degree $\deg(f) = n$ if $a_n \neq 0$, and is *monic* if $a_n = 1$.

Proposition. (The Division Algorithm) Let $f, g \in F[x]$. There exist unique polynomials $q(x), r(x)$, with $0 \leq \deg(r(x)) < \deg(g(x))$, such that

$$f(x) = q(x)g(x) + r(x).$$

Greatest Common Divisors

Definition. $u(x)$ is a *divisor* of $f(x)$ if $f(x) = u(x)v(x)$ for some $v(x)$. $h(x)$ is a *common divisor* of $f(x)$ and $g(x)$ if it divides both.

Proposition. (Euclid's Algorithm) Given nonzero $f(x), g(x) \in F[x]$, there is a unique nonzero monic polynomial $d(x) = (f(x), g(x))$ of minimal degree of the form $a(x)f(x) + b(x)g(x)$. This is the *greatest common divisor*.

1. $d(x)$ is a common divisor of $f(x)$ and $g(x)$;
2. every common divisor $h(x)$ of $f(x), g(x)$ divides $d(x)$.

Roots of the GCD

Proposition. The roots of $d(x) = (f(x), g(x))$ are exactly the common roots of $f(x)$ and $g(x)$. If λ has multiplicity a in f and b in g , its multiplicity in d is the smaller of a and b .

Corollary. $f(x)$ and $g(x)$ have no common root in any algebraic closure of F if and only if there are polynomials $a(x), b(x)$ with

$$a(x)f(x) + b(x)g(x) = 1.$$

Irreducible Polynomials

Definition. $f(x)$ is *irreducible* if it cannot be factored as $f(x) = g(x)h(x)$ with $\deg(g), \deg(h) \geq 1$.

Proposition. If $f(x)$ is irreducible and $a(x)b(x)$ is a multiple of $f(x)$, then either $a(x)$ is a multiple of $f(x)$ or $b(x)$ is.

Unique Factorization

Proposition. A monic polynomial $f(x)$ factors as

$$f(x) = \prod_i p_i(x)^{n_i}$$

where the $p_i(x)$ are distinct monic irreducible polynomials, and the set of p_i and the n_i are unique up to order.

The Primary Decomposition

Let $L : V \rightarrow V$ and factor the characteristic polynomial into distinct irreducibles:

$$P_L(x) = p_1(x)^{n_1} \cdots p_k(x)^{n_k}.$$

Theorem. V decomposes as an internal direct sum

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_k,$$

where $p_i(L)^{n_i} V_i = 0$ for each i .

The Minimal Polynomial

Suppose $\dim V = n$ and $L : V \rightarrow V$ is linear. Cayley–Hamilton gives $P_L(L)V = 0$.

Definition. The *minimal polynomial* $m_L(x)$ is the monic polynomial of smallest degree such that $m_L(L)V = 0$.

Proposition. The minimal polynomial $m_L(x)$ is a divisor of the characteristic polynomial $P_L(x)$.

Cyclic Subspaces

Definition. Let $\dim V = n$. V is *cyclic* for L if there is a $v \in V$ such that $v, Lv, \dots, L^{n-1}v$ span V . Such a v is called a *generator* for V .

If v generates V , there is a relation

$$L^n v = - \sum_{i=0}^{n-1} a_i L^i v,$$

giving a polynomial $q(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ with $q(L)V = 0$. This $q(x)$ is the minimal polynomial of L acting on V .

The Companion Matrix

In the basis $\{L^i v\}_{i=0}^{n-1}$, the matrix of L is

$$A = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & 0 & \cdots & 0 & -a_2 \\ 0 & 0 & 1 & \cdots & 0 & -a_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}$$

Remark. A matrix of this shape for $q(x)$ is called the *companion matrix* of $q(x)$.

The Companion Matrix (cont.)

Proposition. The characteristic polynomial of the companion matrix A is $q(x)$.

Corollary. If V is cyclic for L , then the minimal polynomial and characteristic polynomial of L coincide.

Decomposition into Cyclic Subspaces

Theorem. Let $L : V \rightarrow V$, where $\dim V = n$. Then there is a decomposition

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$$

into cyclic subspaces, where the minimal polynomial on V_i is a power $p_i(x)^{k_i}$ of a monic irreducible polynomial over F . The polynomials $p_i(x)^{k_i}$ are unique up to order.

Two Consequences

- ▶ The minimal polynomial $m_L(x)$ is the least common multiple of the $p_i(x)^{k_i}$.
- ▶ The characteristic polynomial $P_L(x)$ is the product of the $p_i(x)^{k_i}$.
- ▶ Hence every irreducible factor of $P_L(x)$ is a factor of $m_L(x)$.

Proof: Setup

Start with the primary decomposition $V = \bigoplus_i V_i$, where $p_i(L)^{n_i} V_i = 0$, the $p_i(x)$ are distinct irreducibles, and $P_L(x) = \prod_i p_i(x)^{n_i}$.

It suffices to show: if $p(x)$ is irreducible of degree d and $p(L)^n V = 0$, then V splits as a direct sum of cyclic subspaces with minimal polynomials $p(x)^{k_i}$, $\sum k_i = n$.

Proof: Choosing a Generator

Choose $v_1 \in V$ so that $p(L)^{k_1}v_1 = 0$, $p(L)^{k_1-1}v_1 \neq 0$, with k_1 maximal among all such choices. Then $p(L)^{k_1}V = 0$.

v_1 generates a cyclic subspace with basis

$$v_1, Lv_1, \dots, L^{k_1d-1}v_1.$$

These are independent: a dependence relation would give $h(L)v_1 = 0$ for some $h(x)$ of degree $< dk_1$. The gcd $s(x)$ of $h(x)$ and $p(x)^{k_1}$ also satisfies $s(L)v_1 = 0$; since $p(x)$ is irreducible, $s(x) = p(x)^k$ with $k < k_1$, contradicting the maximality of k_1 .

Proof: Constructing the Complement

Let V_1 be generated by v_1 . Goal: find an L -invariant W with $V = V_1 \oplus W$; since W has smaller dimension and satisfies the same hypotheses, induction on $\dim V$ then finishes the decomposition.

Construct W inductively: start with $W_0 = (0)$, and build L -invariant subspaces W_j with $V_1 \cap W_j = 0$, so that for j large enough, $V_j := V_1 \oplus W_j = V$.

Proof: Extending W_j

Suppose W_j is L -invariant with $V_1 \cap W_j = 0$. Choose $w \in V$, $w \notin V_j$.

Since $p(L)^{k_1}w = 0 \in V_j$, and by Euclid's algorithm, the set of $k(x)$ with $k(L)w \in V_j$ is exactly the multiples of some $p(x)^f$, with $1 \leq f \leq k_1$.

Write

$$p(L)^f w = c(L)v_1 + w_j, \quad w_j \in W_j.$$

Proof: Extending W_j (cont.)

Applying $p(L)^{k_1-f}$ to both sides gives

$$0 = p(L)^{k_1-f}c(L)v_1 + p(L)^{k_1-f}w_j.$$

Since $V_1 \cap W_j = 0$, this forces $p(L)^{k_1-f}c(L)v_1 = 0$, so $c(x)$ is divisible by $p(x)^f$: write $c(x) = \delta(x)p(x)^f$.

Set $w' = w - \delta(L)v_1$, let W' be the cyclic subspace generated by w' , and set $W_{j+1} = W_j + W'$.

Proof: $V_1 \cap W_{j+1} = 0$

Suppose $x = a(L)v_1 = w_j + b(L)w' \in V_1 \cap W_{j+1}$. Since

$$b(L)w = a(L)v_1 - w_j + b(L)\delta(L)v_1 \in V_j,$$

we must have $b(x) = p(x)^f \beta(x)$ for some $\beta(x)$. Then

$$b(L)w - b(L)\delta(L)v_1 = \beta(L)(p(L)^f w - c(L)v_1) = \beta(L)w_j,$$

so $x = w_j + \beta(L)w_j \in V_1 \cap W_j = 0$. Hence $V_1 \cap W_{j+1} = 0$.

Proof: Conclusion

Repeating this extension a finite number of times (since $\dim V < \infty$) produces an L -invariant W with $V = V_1 \oplus W$.

Splitting off a cyclic subspace of W and continuing inductively completes the existence proof. (Uniqueness is omitted.)

The Jordan Normal Form

Suppose $P_L(x) = \prod_i (x - \lambda_i)^{n_i}$ factors into linear factors — always true if F is algebraically closed, but not required.

The cyclic decomposition writes $V = \bigoplus_i V_i$, each V_i cyclic with minimal polynomial $(x - \lambda_i)^{k_i}$.

A Basis for a Cyclic Block

Let v_0 generate V_i . The vectors

$$v_j = (L - \lambda_i)^j v_0, \quad j = 0, \dots, k_i - 1$$

form a basis for V_i , and

$$Lv_j = v_{j+1} + \lambda_i v_j \quad (j = 0, \dots, k_i - 2), \quad Lv_{k_i-1} = \lambda_i v_{k_i-1}.$$

In this basis, $L|_{V_i}$ has matrix

$$J_{k_i}(\lambda_i) = \begin{pmatrix} \lambda_i & 0 & 0 & \cdots & 0 & 0 \\ 1 & \lambda_i & 0 & \cdots & 0 & 0 \\ 0 & 1 & \lambda_i & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & \lambda_i \end{pmatrix}$$

Jordan Blocks

Definition. The $k \times k$ matrix with λ on the diagonal, 1 on the subdiagonal, and 0 elsewhere is the *Jordan block* $J_k(\lambda)$.

Theorem. (Jordan Normal Form) If the characteristic polynomial of L factors into linear factors, then there is a basis of V in which the matrix of L is block diagonal, with blocks $J_{k_i}(\lambda_i)$, where $(x - \lambda_i)^{k_i}$ are the minimal polynomials of the cyclic subspaces V_i .

Consequences of Jordan Form

Every matrix A whose characteristic polynomial factors into linear factors is a sum

$$A = D + N$$

where D is diagonal (eigenvalues repeated by multiplicity) and N is nilpotent ($N^k = 0$ for some k).

In a single Jordan block of size k : $\lambda I + N$ with $N^k = 0$ but $N^{k-1} \neq 0$.

Each Jordan block has a one-dimensional space of eigenvectors, so the number of independent eigenvectors equals the number of Jordan blocks.